About Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation, commonly known as RAG, is a technique that combines information retrieval with text generation. Instead of relying only on what a language model learned during training, RAG systems first search a knowledge base for relevant information, then use that information to generate a more accurate and grounded answer.

The RAG pipeline generally consists of three main components: an embedding model, a vector database, and a language model. The embedding model converts text into numerical representations. The vector database stores and searches these representations efficiently. The language model uses retrieved context to generate coherent, relevant responses.

One of the primary benefits of RAG is reducing hallucination, which is when a language model generates plausible-sounding but factually incorrect information. By grounding responses in retrieved documents, RAG systems can provide more trustworthy and verifiable answers, often including citations to the source material.

RAG systems were first popularized in a 2020 research paper by Facebook AI Research (now Meta AI). Since then, RAG has become one of the most widely adopted patterns for building practical applications with large language models, especially in domains like customer support, legal research, and internal knowledge management.

Common tools used to build RAG systems include LangChain and LlamaIndex for orchestration, ChromaDB, Pinecone, and FAISS for vector storage, and OpenAI, Anthropic, or open-source models like Llama for text generation. Evaluation of RAG systems typically involves measuring both retrieval quality and the accuracy of the final generated answer.